

Vivek Yadav Baira

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DevOps Engineer with 4+ years of hands-on experience in automating software deployments, managing build and release processes, and securing infrastructure across Linux and Windows environments.

Professional Summary

- Automated software deployments and managed infrastructure using **Kubernetes**, **Docker**, **Ansible**, and **Chef**, leading to a **40%** reduction in deployment times and improved system reliability. Implemented CI/CD pipelines that streamlined software releases and decreased deployment time.
- Extensive experience with **AWS** and **Azure** platforms, including services like EC2, S3, RDS, VPC, Route 53, and Azure Virtual Machines, Storage Accounts, and AKS. Managed cloud infrastructure using **Infrastructure as Code (IaC)** tools such as **Terraform** and **ARM templates**, ensuring efficient and repeatable deployments.
- Configured **IAM** roles and policies on AWS and Azure, implementing **RBAC**, managing **Firewalls** to enforce the least privilege access, which enhanced security and compliance.
- Proficient in **PowerShell**, **Bash**, and **Python**, automating tasks such as server provisioning, patch management, and deployments. Migrated **ARM templates** from JSON to **Bicep** for improved modularity, and enhancing automation of Azure resource deployment.
- Strong understanding of network architecture and security, including configuring **Azure Virtual Networks**, **VPN** gateways, and load balancers. Managed hybrid cloud environments using **ExpressRoute** and **Azure Traffic Manager**, ensuring high availability and connectivity.
- Managed Kubernetes clusters on AWS and Azure, deploying microservices-based applications that improve system scalability and resilience, resulting in a **30% reduction in downtime**. Used **Helm charts** for consistent deployments across environments.
- Designed and implemented CI/CD pipelines using **Azure DevOps**, **Jenkins**, and **Argo CD**, automating builds, tests, and deployments. Built **Java** applications with **Maven**, managed **Node.js** projects using **npm/Yarn**, handled **Python** dependencies with **pip**. Analyzed code with **SonarQube** and scanned container images with **Trivy**.
- Led database migration from on-premises to AWS, reducing **downtime by 70%** and improving performance. Used **AWS DMS** and **SCT** for data migration, achieving a smooth transition with minimal disruption to business operations and utilized **Amazon RDS** and **AWS DynamoDB** for the new architecture.
- Developed **Python** scripts using **boto3** for **AWS Lambda** to automate the removal of unassociated EC2 snapshots, optimizing resource usage and **reducing costs by 25%**. Configured **CloudWatch** to run the function every hour, optimizing AWS resource usage and achieving cost efficiency.
- Collaborated within **Scrum** and **Kanban** frameworks, driving timely releases and enhancing team productivity by **15%** through streamlined processes. Utilized **JIRA** and **Confluence** to streamline project tracking and documentation, leading to a **20% increase** in team efficiency.
- Managed Oracle and **MySQL databases**, wrote SQL queries for data integrity, and automated migration tasks. Automated **database backups** and recovery processes, reducing downtime from hours to minutes.
- Configured **Splunk**, **Nagios**, **Prometheus** and **Grafana** for monitoring, creating dashboards and alerts that **improved issue detection by 40%**.

WORK EXPERIENCE

Garmin • Olathe, Kansas

06/2024 - Present

Garmin is a leading technology company specializing in GPS navigation, aviation, marine, automotive, and wearable technology, operating in a highly secure and regulated environment.

Azure DevOps Engineer

- Managed **Linux and Windows workloads**, automating system administration, patch management, and package installations using **Bash, PowerShell, and Ansible**. Wrote **runbooks using Azure Automation Accounts** and scheduled jobs for automated cloud operations.
- Automated **Azure infrastructure provisioning** using **Terraform and ARM templates**, implementing a **VCS-driven Terraform workflow** to trigger infrastructure changes through **Git-based version control**, ensuring **auditability, collaboration, and compliance enforcement**.
- Implemented **CI/CD pipelines for applications built in .NET (C#), Java, and TypeScript**, using **Maven for Java builds and .NET build tools for C# applications**. Integrated **Vite for frontend optimization**, **ESLint for static code analysis**, **Docker with multi-stage builds**, **Trivy for vulnerability scanning**, **ACR for storing Docker Image** and **ArgoCD for Kubernetes deployments on AKS**.
- Integrated **SonarQube** in CI/CD pipelines for **Java, .NET (C#) applications**, ensuring **code quality, security scanning, and technical debt analysis**. Managed build artifacts using **Azure Artifacts**, utilized **Azure Repos** for version control.
- Deployed applications on **AKS using Helm charts**, implementing **Istio Service Mesh** for **dynamic service discovery, traffic routing, and improving service-to-service communication reliability**.
- **Designed and implemented Blue-Green and Canary deployment strategies** in **AKS**, reducing downtime and enabling controlled feature rollouts.
- Deployed **Keycloak on Azure Container Apps using the Keycloak Docker image**, integrated it with **Azure AD for Single Sign-On (SSO)**, automated **JWT key rotation**, and implemented **Azure RBAC roles** to enforce least privilege access.
- Defined and configured custom **Azure RBAC roles** for **resource-level access** and implemented **Kubernetes RBAC** for **AKS**, enforcing **least privilege access** and securing credentials with **Azure Key Vault**.
- Developed Azure Functions using C# for event-driven architecture, automating workflows leading to significant cost savings by optimizing resource usage.
- Managed and optimized database solutions using Cosmos DB and SQL Server, enhancing data retrieval performance and storage efficiency.
- **Optimized CI/CD pipeline execution** by replacing AKS with **K3s on GitHub runners** for integration testing, eliminating unnecessary infrastructure dependencies and achieving **cost savings** by removing AKS and Argo CD from Dev environments
- **Implemented monitoring and alerting** for Kubernetes clusters by setting up **Prometheus and Grafana**, configuring **custom dashboards** for real-time visualization, and defining **alert rules in Prometheus Alertmanager** to proactively detect and respond to system anomalies.
- **Enabled distributed tracing for microservices-based applications** by collaborating with developers to instrument code using **OpenTelemetry and Jaeger**, ensuring seamless tracking of request flows across services, optimizing performance bottlenecks, and improving system observability.
- Utilized Azure Service Bus and Durable Functions for event-driven architecture, optimizing workflow automation.
- Designed **Azure Virtual Networks (V-Nets), VPNs, ExpressRoute, Load Balancers, and Network Security Groups (NSGs)** to secure cloud networking. Configured **Azure Web Application Firewall (WAF)** and **F5 BIG-IP** for high-availability proxy setups.
- **Diagnosed and resolved incidents** using **Azure Log Analytics, KQL, and Linux scripting**, improving system stability and reducing downtime.

- Created and maintained **technical documentation for infrastructure, CI/CD pipelines, Kubernetes configurations, and security policies**, enabling **efficient onboarding, troubleshooting, and compliance auditing**. Documented **security incidents, workflows, and operational best practices**

PNC Financial Services • Hyderabad, India

01/2021 - 12/2023

PNC Financial Services is a major American bank holding company providing retail and corporate banking, asset management, and financial services, operating in a highly regulated and security-focused financial environment

DevOps Engineer

- Provisioned **AWS infrastructure** using **Terraform and CloudFormation**, deploying **EC2 instances, RDS databases, EBS volumes, ALB, and Route 53 DNS** to host **highly available application workloads** with auto-scaling and failover mechanisms.
- Configured AWS CloudTrail and created CloudWatch Events to receive SNS Notifications when EC2 instances were terminated. Configured Elastic Load Balancer (ELB) for distributing incoming application traffic across multiple EC2 instances and set up Auto-Scaling groups to handle heavy loads.
- Automated infrastructure deployment in **Azure** using **Terraform**, provisioning **Azure Virtual Machines, VNets, NSGs, Load Balancers, and Virtual Machine Scale Sets (VMSS)** for enterprise workloads and batch processing systems.
- **Built and maintained CI/CD pipelines** using **GitLab CI/CD and GitHub Actions**, leveraging **self-hosted GitHub runners** to optimize builds, reduce pipeline costs, and enable **customized deployment environments** for AWS and Azure workloads.
- **Implemented a 3-tier application deployment on AWS**, configuring **Elastic Beanstalk to automate application hosting, RDS for backend storage, API Gateway for secure API exposure, and S3 for static content delivery**.
- **Managed Docker image lifecycle**, integrating **Docker with Jenkins** to dynamically provision ephemeral build agents, build and package application containers, store artifacts in AWS S3 and Nexus, and clean up containers post-build.
- **Automated application builds** for **.NET (MSBuild), Java (Maven/Gradle), and Node.js**, integrating unit testing, security scanning, and code quality checks in CI/CD workflows
- **Orchestrated containerized workloads on AWS** using **EKS, Helm charts, and Fargate**, ensuring **scalable, highly available, and secure deployments** for microservices applications.
- **Deployed and managed the EFK (Elasticsearch, Fluentd, Kibana) stack on Amazon EKS**, configuring **Elasticsearch as a StatefulSet with EBS persistent volumes using CSI drivers, and integrated IAM roles with the Elasticsearch service account** to securely manage storage access and automate **daily snapshots** for data retention.
- **Implemented CI/CD for databases using ephemeral environments with Neon and GitHub Actions**, automating the creation and deletion of temporary database copies for pull request validation, ensuring isolated schema changes, preventing conflicts in shared development databases, and optimizing cost efficiency
- **Implemented configuration management automation** using **Chef**, provisioning and managing system configurations across AWS and Azure instances, ensuring consistency, compliance, and reducing manual configurations.
- **Configured cloud networking and security** by setting up **AWS Security Groups, Route 53 DNS policies, ELB for traffic distribution** and securing **Azure workloads with VNets, NSGs, and Application Gateway**.
- **Led enterprise data migration from on-prem to Azure**, implementing **Azure Data Factory (ADF) pipelines** to move structured and unstructured data into **Azure Data Lake and Blob Storage**.
- **Implemented centralized logging and monitoring**, integrating **AWS CloudWatch, CloudTrail, and SNS notifications**, while setting up **Log Analytics and Splunk** for log aggregation and security event monitoring.
- **Architected a cost-saving solution on AWS**, developing **Python scripts using Boto3 for Lambda** to analyze **snapshots unassociated with active EC2 instances**, automating the **removal of unnecessary snapshots**. Configured **CloudWatch to run the function every hour**, optimizing **AWS resource usage and achieving significant cost efficiency**.
- Worked with **Agile Methodology** and tools related to code management, build and release automation, and service and incident management.

Labcorp is a global life sciences company specializing in diagnostics, drug development, and healthcare technology solutions, operating in a compliance-driven and data-sensitive environment.

Cloud Engineer

- Maintained source code in **Git**, including branching, creating labels, and merging code for QA testing and releases. Automated various service and application deployments with **Ansible** on **CentOS** and **RHEL**.
- **Automated Linux server configuration** using **Puppet** and **Chef**, creating **custom Cookbooks, modules, and manifests** to ensure **consistent system configurations and compliance**.
- **Provisioned AWS infrastructure** using **CloudFormation**, deploying **EC2, RDS, EBS, ALB, and VPC components** (Subnets, Security Groups, Route Tables, Internet Gateways, Elastic IPs).
- **Designed and managed CI/CD pipelines** using **Jenkins**, automating build, test, and deployment workflows.
- **Designed and implemented multi-region deployments** in **AWS**, ensuring **high availability and disaster recovery** with **Route 53 failover, S3 cross-region replication, and RDS Multi-AZ deployments**.
- **Configured AWS networking** with **VPCs, Security Groups, IAM roles, TLS/SSL certificates, and Apache web server security hardening** using **Puppet automation**.
- **Developed disaster recovery strategies**, implementing **AWS Backup, RDS snapshots, AMI automation, and cross-region replication** to reduce recovery time objectives (RTO).
- **Migrated databases from on-prem to AWS** using **AWS Database Migration Service (DMS), AWS Schema Conversion Tool (SCT), and AWS Snowball** for large-scale data transfers. **Used a combination of lift-and-shift, replatforming, and refactoring strategies** to migrate applications to AWS.
- **Deployed and managed Linux-based workloads**, configuring **CentOS, RHEL, and Ubuntu** on AWS EC2 instances, performing **patch management, security hardening, and system performance tuning**.
- Configured and administered **Network File System (NFS)** to share data between **Linux** and **Windows** environments.
- **Scheduled automation tasks using cron jobs**, managing system backups, log rotation, and application health checks to maintain **uptime and operational stability**.
- Managed and deployed Kubernetes applications using **Helm charts**, created reproducible builds, managed Kubernetes manifest files, and handled Helm package releases. Installed **Docker** and created Docker containers.
- **Implemented cloud monitoring and logging**, installing **Splunk on production servers**, creating **custom dashboards for real-time application performance monitoring**, and setting up **alerts for proactive issue resolution**.
- **Monitored infrastructure health** using **Nagios**, configuring checks for **EC2 instances, network services, and cloud application performance**, ensuring optimal system availability.
- Installed Splunk on production servers for logging purposes, built **Splunk dashboards** for application monitoring, and configured **alerts** for operational purposes. Set up **Nagios** to monitor EC2 Linux instances using Puppet automation.
- **Environment:** AWS, EC2, VPC, S3, RDS, Kinesis, SES, SNS, IAM, Athena, Ansible, Chef, Puppet, SVN, GIT, Java/J2EE, Nagios, Linux, Jenkins, Web Logic, JBOSS, Windows, Linux, RPM.

SKILLS

Operating System: CentOS, Debian Family, Red Hat, Ubuntu, Windows

Cloud Environments: AWS, Azure

Configuration Management Tools: Ansible, Chef, Puppet

DataBase: AWS RDS, Cassandra DB, DynamoDB, MongoDB, MS SQL, MySQL, NoSQL, Oracle, SQL Server

Version Control Tools: GitHub, Mercurial, TVFC

Web Servers: Apache, Tomcat, Web Sphere

Scripting Languages: Bash, Groovy, HTML, Perl, PHP, PowerShell, Python, Ruby, YAML

IaC Tools: Ansible, ARM Templates, AWS CloudFormation, Terraform

Containers and Orchestration Tools: AKS, AWS Fargate, Docker, EKS, Helm, Kubernetes

CI/CD Tools: Aqua Security, Argo CD, Azure DevOps, Checkmarx, GitHub Actions, Gitlab CI/CD, GitOps, Gradle, Jenkins, Maven, MS Build, SonarQube, Synk

Bug Tracking Tools: Azure DevOps, Jira

SDLC: Agile Methodology, Kanban, SCRUM

Artifact Management Tools: AWS CodeArtifact, Azure Artifacts, JFrog Artifactory

Monitoring Tools: AWS CloudWatch, Azure Monitor, ELK Stack, Grafana, Nagios, OpenTelemetry, Prometheus, Splunk

CERTIFICATIONS

Microsoft Certified: DevOps Engineer Expert	09/2024 - 09/2025
https://learn.microsoft.com/en-us/users/vivekyadavbaira-4585/credentials/7cf3044638074ef	
Microsoft Certified: Azure Administrator Associate	09/2024 - 09/2025
AWS Certified DevOps Engineer	

EDUCATION

Master of Science in Computer Science	
University of Central Missouri	01/2023 - 05/2024